

Revised December 1, 2019

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Education

H.B.Sc., Mathematics and Statistics 2018-2022
University of Toronto, Toronto, ON

Technical Skills

Data: Python, Numpy, Pandas, PyTorch, Scikit-Learn, Matplotlib, Seaborn, SQL, NoSQL, R, Excel
Other: shell scripting, vim, version control (git), cloud computing (AWS, Azure, Google Cloud Platform), Unix/Linux, Astropy, Selenium, NLTK, $\text{\LaTeX}$, regular expressions

Relevant Coursework

2019 Fall	Stars and Planets
2019 Fall	Probability and Statistics I
2019 Fall	Ordinary Differential Equations
2019 Summer	Multivariable Calculus
2019 Winter	Introduction to Computer Science
2018-2019	Calculus
2018-2019	Linear Algebra I, II
2018 Fall	An Introduction to Statistical Reasoning and Data Science

Independent Coursework

2019 **Data-Driven Astronomy**, The University of Sydney (Coursera)
Pulsars, black holes, exoplanets, stellar evolution, galaxies. Image stacking, crossmatching, data management, regression, classification. Implementation in Python with Astropy, and SQL.

2019 **Intro to Deep Learning with PyTorch**, Udacity
Mathematics of neural networks. Data preparation. Image classification with CNNs and OpenCV. Time series with RNNs. Implementation in Python with PyTorch.

2018 **Intro to Machine Learning**, Udacity
Classifiers (naïve Bayes, SVMs, ensemble methods), data processing, regression, clustering, outlier detection, PCA, evaluation metrics. Implementation in Python with sklearn.

Projects

2019 **Image classification with CNNs**
Cleaned a dataset of several thousands of images of flowers. Optimized a deep learning model in Python and applied cutting-edge techniques to achieve 99% accuracy in the classification of species of flowers. Could potentially be adapted for classification of astronomical objects.

2018 **Analysis of global Internet usage statistics**
Spearheaded the programming aspect of a collaborative statistical analysis project. Proposed research questions and used R to develop data-backed answers. Gave a presentation of results.

Volunteering

2015-2018 **Python Workshop Lead**, Sir Winston Churchill Secondary, Vancouver, B.C.
Coordinated and delivered workshops to prepare students for a national programming contest.

Interests

Languages: English (native), French, Chinese, Norwegian (conversational)
Other: *nix customization, urban beekeeping, baking, reading, computer programming

Statement of Interest

Use language that shows you are an independent researcher.

- areas of specialty and expertise - academic strengths and abilities - compatibility with the department or school - high ability to think and communicate like a serious scholar and/or scientist

very briefly, who are you? what makes you unique? why are you applying? what abilities do you have? specifically. show dont tell. how can those be useful in this program? how can your research interests be fulfilled by this program?

highlight appealing parts of program.

Discovery and initial characterization of prime targets for future JWST characterization using K2, TESS, and ground-based follow-up

I am excited to be able to apply for this incredible opportunity at the Institute for Research on Exoplanets.

My fascination with space has always been the humility it inspires in me—I find it absolutely incredible that I am alive in this moment, despite life being so unlikely. The study of exoplanets and planetary systems is an exciting area of research which tackle some of the fundamental questions that I am interested in exploring. What conditions observed in our Solar system are unique? How and where are planets formed, and what are they and their atmospheres composed of? How does the evolution of the host star influence those things? And of course, the big question: are we alone? These topics are interesting in themselves, but more importantly, answers to these questions will not provide great implications for not only the origins and evolution of our own Solar System, but also to our greater understanding of the universe. I hope to immerse myself into the exploration of these areas, and I - see no better place to do that than at iRex. - believe that iRex is uniquely positioned to give my the best opportunity to do so.

a paragraph about why irex is awesome

par about why im awesome

The explosion of data and its impact on astronomy are apparent, and being able to understand this information is critical. I have independently sought to develop the skills necessary to do so—as a first-generation student, being able to tackle new problems independently is not foreign to me. I have years of experience programming in Python, with a focus on scientific computing using packages like numpy, pandas, matplotlib, and astropy. I have also learned to leverage the power and versatility of machine learning and deep learning techniques through independent coursework, and have refined those abilities by applying them in independent projects. My programming arsenal, which includes shell scripting, SQL, and cloud computing, allow me to tackle a wide variety of problems. Recently, a course on astronomical computing has taught me the importance of using rigorous techniques in the analysis of data. My academic background in statistics has equipped me with the tools necessary to do so. Perhaps equally as important is being able to communicate, which I believe to be critical to the scientific research process. My experience teaching programming classes, and the numerous report-based projects that I have completed—including a research report I wrote on loss-mass mechanisms in late-stage AGB stars for an astronomy course on stars and planets—have honed my ability to communicate effectively, both verbally and through writing. I am confident in my ability to convey information concisely. These skills position me as a strong candidate to pursue research in this area.

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